An Evolutionary Routing Protocol for Load Balancing and QoS Enhancement in IoT Enabled Heterogeneous WSNs

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Abstract

Energy awareness is a key concern of recent advances in IoT-enabled WSNs, and many optimization approaches to reduce energy consumption have been proposed. The most widely used routing technique for achieving energy efficiency in WSNs is clustering hierarchy. Data transmission over long distances has a negative impact on network efficiency in terms of stability period, network lifetime, and QoS due to the selection of inadequate Cluster Heads (CHs). This paper describes a new Evolutionary Gateway-based Load-Balanced Routing (E-GLBR) algorithm for efficiently selecting appropriate CHs. The proposed algorithm is based on the genetic algorithm optimization method with a new fitness function that takes into account four major parameters to achieve the following goals: improving the CH selection process, reducing energy transmission range, increasing network stability and lifetime, and improving WSN coverage. A comparison simulation with the most recent related methods in MATLAB simulator is performed to evaluate the performance of our proposed algorithm. The simulation results show that our proposed algorithm outperforms all other compared methods in different network scenarios, improving the stability periods of ERP, EEWC, D-MSEP and EDCF by 55%, 43%, 26%, and 12% respectively.

Keywords: HWSNs, Gentic Algorithm, Evolutionary Routing, Gateway-based, Clustering, Load Balancing, QoS Enhancement

1. Introduction

In recent years, wireless sensor networks (WSNs) have emerged as an indispensable and popular generation of IoT environments. WSNs can be defined as groups of hundreds or thousands of sensors of very small size and limited resources, especially in terms of energy supply. These sensors, called nodes, are typically, deployed to measure physical phenomena such as temperature, pressure, vibrations, sound, presence of objects, light intensity, seismic movements, etc. [1, 2, 3, 4, 40]. A wireless network is the result of the fusion of two fields of modern computing: embedded systems and wireless communications. Network sensors are deployed in a coverage

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area to collect the information that comes from an occurring event, the collected data is communicated between sensor nodes and the user via wireless links [4, 5]. WSNs are used today in diverse fields to perform a wide range of tasks including environment monitoring, earthquake detection, military applications, healthcare systems, and scientific exploration, and more.However, there are plenty of challenges facing this generation of networks, including data delivery, efficient use of nodes energy, security, scalability, robustness, connectivity and coverage. [40, 5, 6, 7]. Because of their small size and low cost, network nodes are very limited, particularly in terms of power resources. Each sensor node is equipped with a small battery that supplies the sensor with energy to perform various tasks such as data collection, processing and transmission. Each sensor node is involved in three main tasks, data collection, information processing and transmission. Most of node energy is consumed in the transmission phase. After their deployment, sensor nodes are left untended, nodes batteries replacement or recharging is practically impossible, when a sensor node consumes all its energy, it is considered dead and the network loses one agent in the covered area[8, 9, 10, 11]. The most research works in the field of WSNs applications are directed towards increasing the lifetime of the network. The proposed methods are generally based on the clustering hierarchy where sensor nodes are divided to form some clusters to eliminate or reduce the unnecessary energy waste [5, 12, 4]. Each cluster is led by one node as cluster head (CH), the central role of each CH is aggregating other cluster members data and forward it to the base station. Routing protocols, which are based on clustering hierarchy, have shown great efficiency in extending the lifetime of the network and improving its throughput, but cluster head selection still presents another source of energy drain, especially when the selection is not efficient in covering all network nodes or only a few nodes are chosen as cluster heads during all network operations, which causes a reduction in network throughput due to the loss of many nodes in a few rounds after being CH. To handle this issue, many researchers have worked on the optimization of CH selection using various selection strategies and optimization methods. The selection of adequate CHs is defined as a non-deterministic polynomial-time hard (NP-Hard) problem. Observation of nature has led researchers to borrow the principles observed in natural phenomena to create efficient algorithms for solving certain difficult problems including the NP-Hard problems. A new era is opened with bio-inspired algorithms which are meta-heuristics optimization methods [13, 14].

The basic concept of most metaheuristic algorithms is inspired by nature, animal behavior or physical phenomena. Evolutionary methods mimic the evolutionary process in nature to find efficient solutions for complex optimization problems [13, 15, 16]. Metaheuristic optimization methods have led to the discovery of a large number of optimization algorithms in various categories, such as Evolutionary algorithms, Algorithms based on Swarm Intelligence, Physics-based optimization algorithms [13, 17]. Evolutionary algorithms generally use an iterative procedure, based on a progression of biological evolution to solve optimization problems. From such a large solution space, the Genetic Algorithm (GA) is one of the most suitable methaheuristics for optimal routing and efficient load balanced clustering.

In this paper, we propose a new algorithm for CH selection by incorporating significant parameters within a new fitness function developed for the GA evolutionary optimization technique. The rest of this paper is organized as follows: Section 2 presents the literature review of traditional clustering techniques and evolutionary routing protocols. In Section 3, we explain the deployment of the heterogeneous WSN model with the problem formulation. Section 4 discusses the details of the proposed developed fitness function. Section 5 concerns the simulation results and discussions. Section 6 concludes with final remarks and future works.

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2. Literature Review

WSNs applications require efficient use of energy, security, scalability and efficient data routing that ensures a good quality of service. Routing protocols are differently designed to fit this requirements. The choice of a routing protocol is quite large, in this section we classify them into traditional routing protocols such as LEACH [18], SEP [19], and DEEC [20], and evolutionary routing algorithms like HCR [21], ERP [22], and EEWC [23].

2.1. Traditional routing Protocols

The LEACH (Low Energy Adaptive Clustering Hirerachy) protocol divides the network into clusters with a random selection of the cluster heads, this sort of selection does not care about the residual energy of the selected sensor node. This can lead to early death of the chosen node, and as a result, the lifetime of the node is reduced. One other issue with LEACH is that the chosen cluster heads may not cover the entire network because they are chosen randomly. The centralized LEACH (LEACH-C) [24] solves these problems. This protocol counts on a central control algorithm to disperse cluster heads all over the network and enhance the network coverage. The selection of cluster heads and formation of clusters is done by the base station, each cluster head selection is done based on the residual energy of the concerned node, therefore, in each round, each node is required to send the information about its energy level to the base station that in turn shall decide whether that node is eligible to become a cluster head for the current round or not. A two-level heterogeneous Stable Election Protocol (SEP) was proposed in [19]. In the SEP scheme, there are two types of sensor nodes: normal and advanced nodes with two different energy levels. Advanced nodes are fewer in number than normal nodes due to their high cost, but they carry a higher amount of initial energy. However, in an attempt to prevent normal nodes from having high probabilities of becoming cluster heads, which can deplete their energy in a short period of time, SEP uses two election probabilities weighted by the type of node. The main drawback of the SEP protocol is that it cannot scale to multilevel heterogeneous environments that hold of more than two energy levels. To deal with heterogeneous WSNs with three energy levels, Singh and Malik [27] proposed the Heterogeneous SEP protocol (hetSEP) for increasing lifetime in WSNs. The proposed hetSEP scheme decreases the network energy consumption rate through an improvement in the weighted election probability discussed in [19]. Depending on a single parameter for CH selection, this protocol can only handle one level, two levels and three levels of energy heterogeneity, therefore it cannot handle multi-level heterogeneous WSNs. With the aim of enhancing network lifetime and prolong the stability period in two-level heterogeneous WSNs, two improving versions of SEP, known as Modified Stable Election Protocol (M-SEP) [28] and Distance Incorporated Modified Stable Election Protocol (D-MSEP) [29] have been proposed.M-SEP integrates the residual energy of the node and the average energy into the CH threshold election formula. D-MSEP adds the distance factor to this threshold to avoid transmission over a long distance. In order to reduce the power consumption rate and extend the network lifetime in heterogeneous wireless sensor networks with multiple energy levels, a Distributed Energy Efficient Clustering (DEEC) protocol was developed in [20]. For energy load balancing, DEEC scheme rotates the cluster-head role among all nodes. The ratio between the residual energy of each node and the average energy of the whole network is used to compute the CH selection probability more efficiently. Every single node has its own probability to be a cluster-head, this means that each node has a different number of cluster head rounds depending on its initial and residual energy. This approach ensures that nodes with more remaining energy are more likely to be cluster heads than nodes with less remaining energy. Some DEEC

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protocol enhancements were proposed in Distributed DEEC (TDEEC) [30], Enhanced Developed DEEC (EDDEEC) [31] and Developed DEEC (DDEEC) [32]. The main contribution of TDEEC algorithm was enhancing the stability period and reduce the energy consumption rate by modifying of the CH selection threshold value. In the new proposed threshold only nodes with higher residual energy are eligible to be cluster heads. In EDDEEC, the authors introduced a new scheme where the cluster head election probability is dynamically changed according to an absolute energy level threshold. The developed threshold ensures that all nodes have the same probability of being CH when they have almost the same residual energy, this way nodes with higher initial energy are prevented from being cluster heads in a large number of rounds compared to other nodes. With the same goal of preventing nodes with a higher election probability from being CH for a large number of rounds, the DDEEC scheme is based on a residual energy threshold that ensures that a node is eligible to be CH only if it meets the energy threshold requirement. under the value of the predefined threshold, all nodes are eligible to be CHs. With the aim of improving the network lifetime of DEEC protocol in 3-level heterogeneous WSNs, Singh et al. [33] described a heterogeneous DEEC (hetDEEC) protocol that uses a new parameter in the CH election probability to extend the network life of 3-tier heterogeneous DEEC. HetDEEC performed similarly to DEEC in both 1- and 2-level heterogeneity. The Energy Dissipation Forecast and Clustering Management (EDFCM) protocol has been discussed in [34] for two-level heterogeneous networks. When selecting cluster heads, the EDFCM scheme considers the residual energy and energy consumption rate of all sensor nodes. Cluster heads are chosen using a mathematical model that predicts one-step energy usage and ensures the best number of cluster heads. In each loop of the network, the number of CHs remains constant. This is the protocol's key advantage, although the CH selection method lacks information on cluster size and location. A three-level heterogeneous clustering protocol (HCP) was discussed in [35] for WSN lifespan enhancement. The proposed heterogeneous network is divided into three groups, with residual energy and network average energy factored into the threshold election factor to select the best node for cluster head. Energy dependent cluster formation (EDCF)[36] is another method for extending network lifetime in two-tier energy WSNs has been proposed. The authors extended the traditional threshold function used in cluster head selection to include node energy; a sensor node can become a CH only if it meets the threshold requirements, depending of its remaining energy.

2.2. Evolutionary routing protocols

The routing methods described above cannot always guarantee efficient CH selection since they use only a few parameters in the selection process. To improve network performance by selecting efficient CHs, it is necessary to move towards evolutionary routing protocols where the inclusion of multiple parameters is allowed. in HCR [21], the clustering process is managed by the base station which it uses a genetic algorithm (GA) to create energy-efficient hierarchical clusters. To enhance network throughput, the GA fitness function is developed to include more parameters such as the estimated transmission energy, standard deviation of the cluster distance and the number of transmissions. The generated GA-based clusters are broadcast to sensor nodes to configure the network accordingly. Bara et al. [22] proposed an evolutionary-based routing protocol (ERP) with a modified fitness function to prolong the network lifespan. The fitness function of the proposed scheme incorporates two clustering parameters which are the cohesion and separation error. GADA-LEACH [37] is another genetic algorithm based protocol with more awareness of transmission distance. The protocol aims to reduce the energy dissipation of the

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clusters by improving the cluster head (CH) selection process. The fitness function of GADA-LEACH includes the distance between the cluster head and its cluster members, the CHs to the BS distances and nodes energy parameter. An Heuristic Algorithm for Clustering Hierarchy (HACH) is introduced in [37]. The proposed protocol operates with three consecutive phases: sleeping mode schedule, clustering process and network operations. At the beginning of each round, inactive nodes are sent to sleeping mode to prevent the unwarranted energy waste. To achieve a better CH nodes distribution, a new crossover operator is incorporated in the clustering process. The fitness function is also developed by combining a predefined risk penalty parameter and the ratio between the total energy of all non-cluster head nodes and the total energy of cluster head nodes. Energy-efficient weighted clustering (EEWC) [23] is also a genetic-algorithm based method. EEWC protocol modifies the set-up phase of LEACH protocol using the genetic algorithm. For more efficiency in the clustering process, the authors proposed a new objective function that consider three main clustering parameters which are: number of cluster heads, compactness and separation error. A new version of the adaptive low-energy clustering hierarchy with genetic optimization algorithm (O-LEACH) is presented in [38], to prevent the network nodes from consuming a lot of energy in communicating with the CHs, the O-LEACH algorithm ensures the selection of enough CHs to cover the entire network. The objective function of this algorithm takes into account the residual energy of the nodes as the main parameter in the CH selection process. The implementation of micro genetic algorithm in LEACH protocol is discussed in [39], the selection of CHs is improved by relying on GA selection algorithm instead of probability threshold selection technique. To avoid data transmission over long distances, a relay node is introduced between the farthest CHs and the base station.

3. Heterogeneous WSN model and problem formulation

This section describes the details of the heterogeneous network model and the proposed enhanced GA for cluster head selection.

3.1. Network deployment

For network nodes deployment, on count on the model proposed in [40]. Considering N wireless sensor nodes distributed in a square $M \times M$ field.Nodes are supposed to remain stationary when the network is deployed, and each sensor node is equipped with a global positioning system (GPS) that facilitates in determining its location. Battery replacement is not possible since sensor nodes are left unattended after deployment. All sensor nodes have comparable processing and communication capabilities. The network's heterogeneity is assessed in terms of node energy, which varies from node to node. The base station, which is regarded to be outside the network, receives data from all sensor nodes at all times. The discovered data is first collected by a gateway in the network's center, which then forwards it to a base station outside the network. The system has no energy, processing, or memory constraints. For data communication, the radio links between all nodes are assumed to be symmetric, thus data transmission from node i to node j consumes the same amount of energy as data transmission from node j to node i.

The transmission distance has a significant impact on the lifetime of sensor nodes and the energy consumption rate of the entire network. To minimize rapid depletion of network energy, sensor nodes should avoid communicating with other nodes over large distances. For that reason, the monitored field has been separated into four zones, as illustrated in Fig.1. Nodes in the direct transmission zone Z_D are so close to the base station, they will communicate their data without

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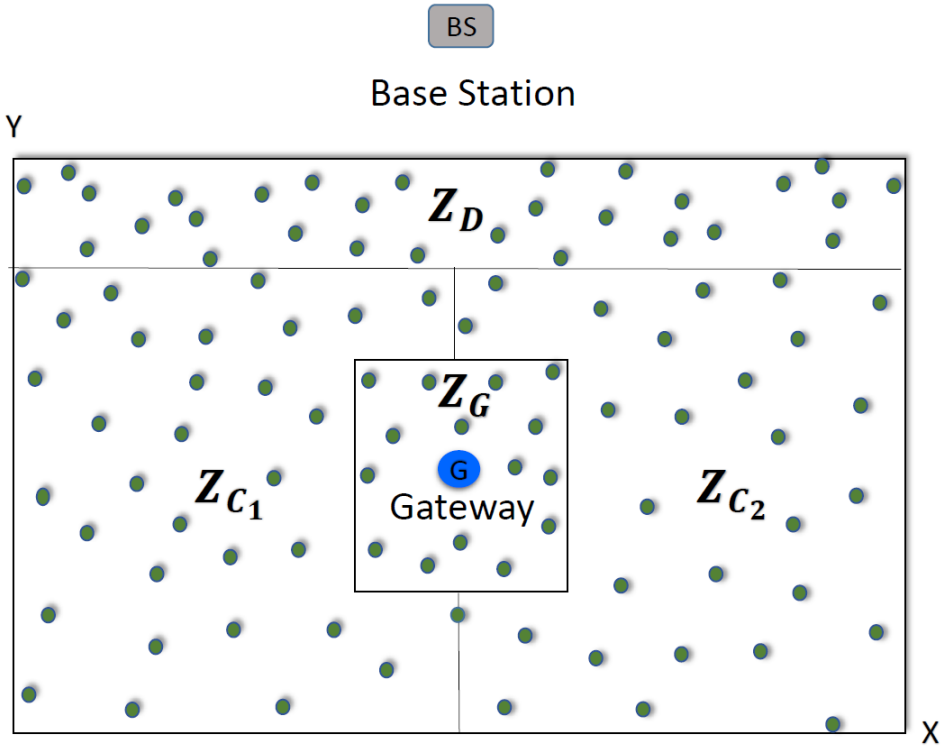

Figure 1: Network model: Nodes in zone Z_D (direct communication zone) send data to the base station directly. The gateway receives data from the nodes Z_G. Nodes in clustered fields $Z_{C_{1,2}}$ choose selected cluster heads to transfer their data to the gateway, which then delivers it to the base station. [40].

the need for any intermediate nodes. The gateway collects data from nodes in the gateway zone ZG before forwarding it to the base station. The collected data is communicated to the base station via the intermediate of some elected cluster heads in the two clustered zones $Z_{C_{1,2}}$.

2-tier heterogeneity

In 2-tier energy heterogeneity, two kind of nodes are placed in the monitored field, $(1-m)*N$ normal nodes equipped with $E_{nrm} = E_0$ as initial energy and $m * N$ advanced nodes equipped with $E_{adv} = E_0 * (1 + \alpha)$, which is α more times than the lowest energy E_0, the total energy of the whole network is:

$$E_{total} = E_0 * N * (1 + m * a) \tag{1}$$

To describe the energy dissipated in a sensor node while it is communicating, we use the usual first radio model where the energy dissipated to transmit a $L - bit$ message is given as:

$$E_{T_x}(L, d) = \begin{cases} L * E_{elec} + L * \epsilon_{fs} d^2 & \text{if } d \leq d_0 \\ L * E_{elec} + L * \epsilon_{mp} d^4 & \text{if } d > d_0 \end{cases} \tag{2}$$

where E_{elec} is the energy dissipation per bit in a transceiver circuit, d the distance between the transmitter and the receiver, ϵ is the amplification energy and d_0 presents a threshold distance

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calculated by:

$$d_0 = \sqrt{\frac{\epsilon_{fs}}{\epsilon_{mp}}} \tag{3}$$

The required energy to receive a $L - bit$ message,is given as follows:

$$E_{R_x} = L * E_{elec} \tag{4}$$

For data aggregation, the consumed energy by each CH is:

$$E_{agg} = CM * L * E_{DA} \tag{5}$$

Where CM presents the number of nodes that belong to the cluster and E_{DA} is the required energy to aggregate one bit of the received data.

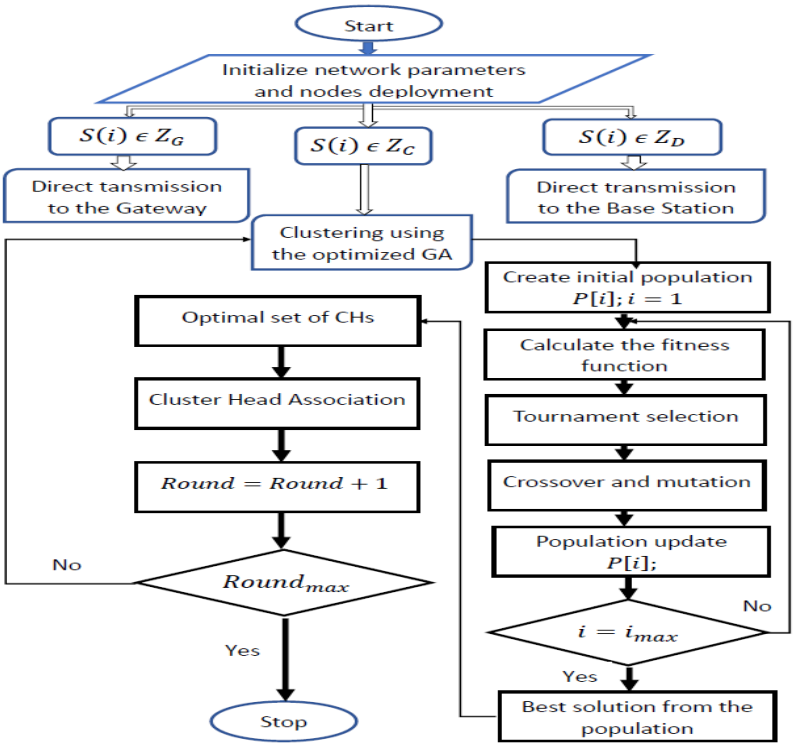

Figure 2: The overall flowchart of the proposed algorithm

4. Proposed Evolutionary Routing Protocol

In this section we discuss the details of the developed new fitness function of the proposed Evolutionary Gateway-based Load-Balanced Routing protocol (E-GLBR). Fig.2 illustrates the overall flowchart of the proposed E-GLBR algorithm

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4.1. Initial Phase

After the deployment of all sensor nodes, each node sends to the BS an ID table contains the location, residual energy of a node and its distance to the Gateway and base station.

4.2. Cluster-head election

At the beginning of each round, the proposed algorithm selects some cluster heads to aggregate the collected data in the two clustered zones, the selection process starts with creating a population of individuals where the individual is presented by a fixed-length list with a size equal to the number of sensor nodes in the clustered-zone. A normal node is presented by 0 and the cluster head is presented by 1. When a sensor node is dead, it will be presented by -1. At the first round, each node is initialized by 0 or 1 according to a predefined election probability p :

$$\forall i \in [1, \ldots, n] \, and \forall j \in [1, \ldots, N_{1,2}] \tag{6}$$

$$I_j^i = \begin{cases} 1 & \text{if } E(node_j) > 0 \quad and \quad random_j \leq p \\ 0 & \text{if } E(node_j) > 0 \quad and \quad random_j > p \\ -1 & otherwise \end{cases} \tag{7}$$

To evaluate the goodness of each individual, we reconsidered the fitness functions proposed by the evolutionary methods discussed in section 3, our new fitness function counts on four main factors including nodes residual energy, transmission distance, nodes density, and CH energy consumption.

4.2.1. Residual energy factor

Due to the additional responsibilities tied to the cluster head including data aggregation and the communication with the gateway, being a cluster head exhausts CHs nodes energy as compared to non-CH nodes which are limited just on node to node data transmission. To avoid the quick death of a particular type of nodes, the selection of CH should consider the remaining energy of each node. Sensor nodes with higher remaining energy should be the ones which own more chances to be elected as CHs. Hence, the first parameter to be considered in our fitness function is the residual energy factor described as follows :

$$F_1 = \sum_{i=1}^{i=N_{1,2}} \frac{E_{res}(i)}{E_{ini}(i)} \tag{8}$$

4.2.2. Transmission distance factor

The transmission for long distances depletes nodes energy so fast which leads to a decrease in the network lifetime. The CH-Gateway transmission distance is an important parameter to be considered in the CH selection process, nodes which are so close to the gateway are more adequate to be CHs. The wider the communication distance, higher the consumed energy and vice versa. Thus, the fitness function should consider the transmission distance parameter to ensure the shortest possible data transmission distances.

$$F_2 = \left(\sum_{i=1}^{i=N_{1,2}} \frac{Dist_{S-G}(i)}{MaxDist_{S-G}} + \frac{1}{AvgDist_{S-G}} \right) \tag{9}$$

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Where, $Dist_{S-G}$ signifies the distance between a sensor node and the gateway and $MaxDist_{S-G}$ indicates the distance between the gateway and the farthest node. $AvgDist_{S-G}$ is the average distance between all nodes and the gateway, this parameter is included to enforce the selection of the nodes nearest to the gateway position as CHs.

4.3. Nodes density factor

A typical CH is the node that is surrounded by a maximum number of more nearby cluster members than the other CHs. Minimizing the intra-cluster distance has a major impact on reducing cluster nodes energy consumption rate. Hence, nodes density factor is an important factor that should be minimized :

$$F_3 = \sum_{j=1}^{N_{C_{1,2}}} \sum_{i=1}^{N_{1,2}} \frac{Dist_{i-j}}{Min_{CHs_{dis}}} \tag{10}$$

Where, $Dist_{i-j}$ is the distance between an ordinary node and a CH, and $Min_{CHs_{dis}}$ is the minimum the distance between two CHs.

4.3.1. CHs energy consumption factor

The selection of a node with low residual energy causes the quick death of this node in a few rounds after being a CH. In each round some cluster heads should be selected to guarantee data communication between sensor nodes and the gateway, if many nodes are disappeared after being CHs, the network falls in the hot-spot problem where many locations of the monitored field may get isolated of the network. To insure the selection of nodes with minimum energy consumption rate, we include the factor of CHs energy consumption described as follows :

$$F_4 = \sum_{i=1}^{N_{C_{1,2}}} \frac{E_{CH}}{E_{res}(i)} \tag{11}$$

4.3.2. Fitness function

Our objective function is to balance the energy load over all sensor nodes and reduce the energy consumption rate of the whole network. The discussed work is directed to the sense of minimizing the proposed fitness function to achieve an optimal network performance. Therefore, our fitness function is constructed as follows :

$$Fitness(Chrom) = \alpha * \frac{1}{F_1} + \beta * \frac{1}{F_2} + \gamma * \frac{1}{F_3} + \sigma * \frac{1}{F_4} \tag{12}$$

Where α, γ, β, and σ are predefined wights respectively assigned to nodes residual energy, transmission distance, nodes density, and CH energy consumption factors. The following algorithm describes all steps in the CH selection process.

5. Simulation Results

To evaluate the performance of the proposed work, a simulation model was developed to investigate the efficiency of our algorithm against Evolutionary Routing Protocol (ERP), Energy Dependent Cluster Formation (EDCF), Distance incorporated Modified Stable Election Protocol (D-MSEP) and Energy-Efficient Weighted Clustering method (EEWC). The testing parameters

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Algorithm 1 CH selection steps

Require: $Set\ of\ nodes\ N_i,\ nodes\ IDs,\ S(i),\ PopulationSize, MaxIteration,\ P_c,\ P_m,\ \alpha,\ \beta,\ \gamma,\ \sigma,\ ElectionProbability\ p$

```
Initilizing Population Seeds
  Chromosome = zeros(PopulationSize, N_i)
  Fitness = zeros(1, PopulationSize)
Initialize Population
for i = 1 : 1 : PopulationSize do
    for j = 1 : 1 : N_i do
        if S(i).Energy ≤ 0 then
            Chromosome(i, j) = −1
        else
            if random_j ≤ p then
                Chromosome(i, j) = 1
            else
                Chromosome(i, j) = 0
            end if
        end if
    end for
end for
Fitness Calculation and Evaluation
  Calculate F_1, F_2, F_3, and F_4
  [Chromosome, Fitness] = CalculateFitness(PopulationSize, N_i, S, Chromosome, Fitness)
for i = 1 : 1 : MaxIteration do
    [Chromosome, Fitness] = Selection(PopulationSize, N_i, Chromosome, Fitness)
    [Chromosome, Fitness] = Crossover(PopulationSize, P_c, N_i, Chromosome, Fitness)
    [Chromosome, Fitness] = Mutation(PopulationSize, P_m, N_i, Chromosome, Fitness)
    [Chromosome, Fitness] = CalculateFitness(PopulationSize, N_i, S, Chromosome, Fitness)
end for
Find the best individual with minimum fitness
  MinFitness = Fitness(1);
for i = 0 : 1 : PopulationSize do
    if Fitness(i) ≤ MinFitness then
        MinFitnessIndividual = i
        MinFitness = Fitness(i)
    end if
end for
```

are given in Table.1. The mentioned routing protocols are implemented in Matlab. The simulation is performed with two-tier energy heterogeneity; 20 advanced nodes and 80 normal nodes are distributed in a $100*100m^2$ field, the base station is located out of the network. To investigate the effectiveness of our proposed protocol, we consider the following performance metrics: *stability period*, *nodes lifetime*, *energy consumption rate*, *network coverage and number of packets sent to th BS*.

5.1. Stability period

The stability period is the duration from the start of the network operation until the death of the first node (FND), it plays a crucial role to evaluate the reliability of the network. As shown in figure tables 2 and 3 the results obtained by our proposed algorithm outperform those

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Table 1: Network parameters

Description	**value**
Monitored field size	$100 \times 100m^2$
Number of nodes N	100
Energy to transmit at a shorter distance ϵ_{fs}	$10\ nJ/bit/m^2$
Initial energy of a normal node E_0	$0.5\ J$
Energy to transmit at a longer distance ϵ_{mp}	$0.0013\ pJ/bi/m^4$
Energy consumed in the electronics circuit E_{elec}	$50\ nJ/bit$
Energy for data aggregation E_{DA}	$5\ nJ/bit/signal$
Base station position	$(50\ m, 120\ m)$
Percentage of advanced nodes m	20%
Sensing range radius R_s	$5.64\ m$
Population size P_s	5
Crossover rate P_c	0.6
Mutation rate P_m	0.002
Crossover type C_t	Two point crossover
Selection type S_t	Tournament selection
Packet size L	$4000\ bits$

of ERP, EDCF, D-MSEP and EEWC. In E-GLBR protocol, the stability period is prolonged in all values of m and a. Nodes in ERP, EDCF, D-MSEP and EEWC exhaust their energies in a very short period of time. Fig.3 illustrates the stability period for different values of m and α. In the scenarios where the percentage of advanced nodes (m) is varying, the stability period is limited at 1080 rounds for EEWC, 1437 rounds for D-MSEP, 1086 rounds for ERP and 1438 rounds for $EDCF$, while our proposed algorithm preserves all nodes energy for more than 1600 rounds in all scenarios, the maximum value of the stability period reached by our proposed E-GLBR is 1656 rounds which presents 576 rounds more than EEWC, 570 more than ERP, 219 rounds higher than D-MSEP and 218 rounds than EDCF. For the other scenario where the energy percentage α is varying, our proposed algorithm can still operate for more rounds than the other protocols, the maximum stability periods achieved by all networks are: 1084 for ERP, 1106 for EEWC, 1458 for D-MSEP, 1494 for EDCF and 1647 rounds for our E-GBLR.

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Table 2: Stability period under 2-level heterogeneous model when m is varying.

Values of m	**Routing Protocols**	*FND*	10% **dead**	20% **dead**	30% **dead**	40% **dead**	50% **dead**	60% **dead**	70% **dead**	80% **dead**	90% **dead**	100% **dead**
	ERP	1046	1190	1253	1298	1346	1396	1437	1499	1591	1896	6580
	EDCF	1396	1553	1679	1770	1855	1952	2057	2188	2328	2472	11045
m =0.1	D-MSEP	1305	1510	1617	1690	1777	1868	1951	2045	2147	2308	8260
	EEWC	1069	1214	1294	1385	1423	1479	1532	1605	1687	1942	3994
	E-GLBR	1600	2057	2335	3405	5381	6319	6695	7062	7457	7699	9740
	ERP	1058	1206	1273	1320	1374	1437	1490	1587	1856	5176	6958
	EDCF	1358	1619	1760	1845	1953	2071	2208	2360	2562	8886	11373
m =0.2	D-MSEP	1372	1521	1586	1636	1681	1750	1802	1863	1995	5301	6657
	EEWC	1045	1210	1310	1378	1443	1508	1588	1716	2023	4391	5167
	E-GLBR	1641	2152	2685	3517	5350	5825	6137	6448	6804	7159	9428
	ERP	1064	1214	1277	1345	1412	1480	1562	1870	4861	5388	6690
	EDCF	1426	1659	1781	1895	2018	2150	2346	2568	8198	9397	11561
m =0.3	D-MSEP	1437	1621	1715	1787	1857	1951	2071	2222	5246	5810	7153
	EEWC	1032	1244	1333	1413	1469	1556	1666	2050	4381	4939	5455
	E-GLBR	1656	2169	2622	4386	4963	5470	5849	6092	6326	6976	9286
	ERP	1066	1226	1309	1375	1451	1559	1871	4783	5129	5600	6934
	EDCF	1424	1683	1853	1992	2131	2327	2560	8083	8993	10175	11750
m =0.4	D-MSEP	1410	1642	1736	1812	1913	2013	2175	5081	5610	6031	7083
	EEWC	1080	1271	1367	1449	1544	1661	2048	4433	4883	5296	5766
	E-GLBR	1627	2221	2729	4165	4800	5158	5578	5818	6027	7810	9398
	ERP	1086	1248	1336	1427	1537	1877	4754	5041	5300	5713	6887
	EDCF	1438	1714	1912	2129	2305	2556	7635	8364	9173	10270	11595
m =0.5	D-MSEP	1413	1680	1811	1951	2074	2216	5160	5582	5956	6376	7382
	EEWC	1072	1284	1407	1509	1668	1947	4408	4837	5230	5538	5915
	E-GLBR	1624	2283	3043	4073	4645	5095	5323	5510	5855	8083	9278

Table 3: Stability period under 2-level heterogeneous model when α is varying.

Values of α	**Routing Protocols**	*FND*	10% **dead**	20% **dead**	30% **dead**	40% **dead**	50% **dead**	60% **dead**	70% **dead**	80% **dead**	90% **dead**	100% **dead**
	ERP	1043	1221	1298	1349	1427	1523	1612	1744	1886	2069	2540
	EDCF	1474	1772	1899	2023	2147	2319	2586	2919	3281	3798	4539
α =0.5	D-MSEP	1242	1502	1591	1672	1779	1903	2011	2151	2355	2597	3021
	EEWC	1013	1230	1343	1428	1502	1593	1708	1844	2009	2136	2343
	E-GLBR	1645	2168	2345	2449	2819	3322	3626	3859	4157	4413	4630
	ERP	1046	1219	1298	1354	1434	1587	1786	2066	2333	2706	3592
	EDCF	1437	1672	1826	1948	2072	2209	2702	3250	3853	4615	5946
α =1	D-MSEP	1366	1537	1603	1653	1709	1793	1935	2162	2437	2744	3577
	EEWC	1065	1235	1330	1417	1510	1608	1872	2167	2417	2656	2913
	E-GLBR	1643	2155	2488	2884	3678	4089	4275	4431	4676	4953	5313
	ERP	1063	1214	1282	1354	1431	1549	1891	2317	2729	3281	4542
	EDCF	1494	1704	1895	1988	2135	2274	2919	3677	4554	5613	7660
α =1.5	D-MSEP	1386	1617	1734	1841	1943	2047	2491	3000	3631	4097	5590
	EEWC	1059	1279	1361	1434	1517	1643	1943	2302	2742	3207	3714
	E-GLBR	1630	2131	2381	3138	4118	4418	4576	4744	4926	5255	6218
	ERP	1078	1201	1276	1341	1415	1554	2046	2594	3110	3864	5184
	EDCF	1421	1632	1774	1923	2039	2221	3123	4032	5257	6733	9213
α =2	D-MSEP	1414	1585	1699	1823	1963	2113	2712	3327	3908	4899	6701
	EEWC	1073	1257	1342	1416	1491	1611	2102	2600	3030	3643	4237
	E-GLBR	1575	2210	2810	3990	4734	4939	5209	5506	5945	6474	7465
	ERP	1084	1229	1289	1353	1405	1512	2157	2794	3541	4461	6443
	EDCF	1404	1643	1797	1935	2057	2243	3298	4443	5799	7766	10587
α =2.5	D-MSEP	1458	1616	1685	1741	1788	1849	2350	2864	3302	3883	5536
	EEWC	1106	1253	1341	1413	1517	1657	2271	2840	3474	4170	4902
	E-GLBR	1647	2223	2801	4346	5281	5596	5767	5903	6114	7024	8492

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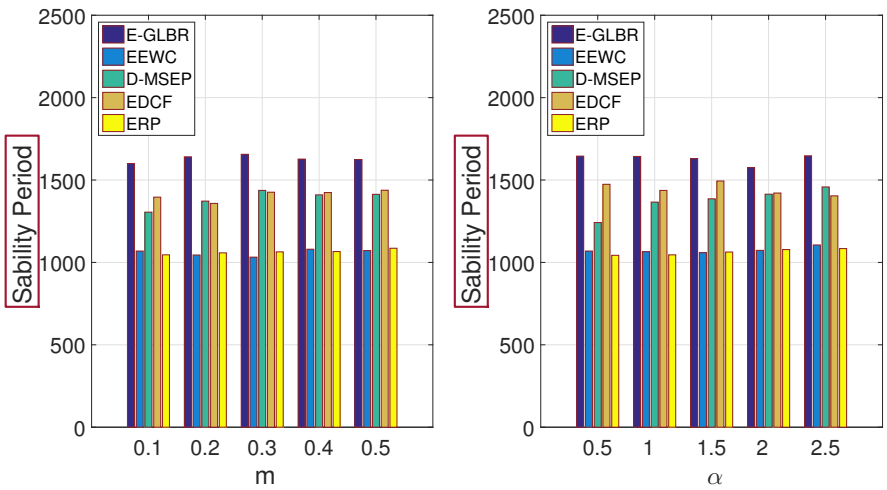

Figure 3: Stability periods in mean cases for different values of m and α.

5.2. *Network lifetime*

The network lifetime metric counts the number of rounds from the start of network operation until the death of the last node. As demonstrated in Fig.4, in our proposed E-GLBR the

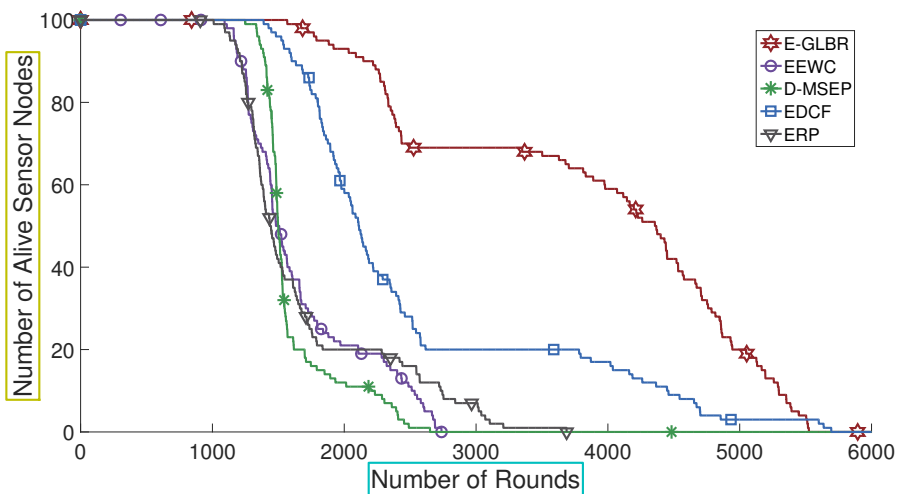

Figure 4: Alive nodes Vs. rounds for $m = 0.2$ and $\alpha = 1$.

last node depletes its entire energy after 5525 rounds which presents 201%, 208% and 149% the network lifetime of EEWC (2738 rounds), D-MSEP (2651 rounds) and ERP(3687 rounds) respectively.The longest lifetime is achieved by EDCF (5695 rounds) which is higher than the lifetime of our algorithm, but the crucial point here is that in the last 3000 rounds, EDCF network keep operating just with 20 nodes, which presents just 20% of the total number of network

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nodes, nodes keep sensing and sending their data without reaching the gateway and the base station which will have a huge effect on the network throughput and coverage. In the same last 3000 rounds, in our proposed protocol only 30 nodes stop working and the network still has 70 nodes to operate in the next rounds, this results is due to the limitation of the transmission distance and the crucial role of the gateway which reduces the energy load and the transmission range by aggregating and forwarding CHs nodes data to the base station, contrary to the other protocols where the CHs can communicate directly with the base station which allows the continuation of the simulation until all the nodes forming the network are dead.

5.3. Energy consumption rate

Fig.5 illustrates the energy consumption rate of all compared methods. Each network starts its operation with $60J$ of initial energy. The energy dissipation rate of D-MSEP, ERP and EEWC protocols is much higher which leads to the quick death of networks nodes, in less than 1500 rounds 83% of networks energy is consumed, at this round EDCF and E-GLBR networks still has respectively 38% and 58% of their initial energy. At the 3000 round D-MSEP, ERP and EEWC networks are almost dead, EDCF network still has just 10% of its total initial energy to operate in the next rounds, the network consumed all its initial energy in 5695 rounds. After 3000 rounds, the proposed E-GLBR still owns more than 25% of its initial energy, which gives the network more rounds to operate, the last amount of network residual energy is consumed at the 5525 round.

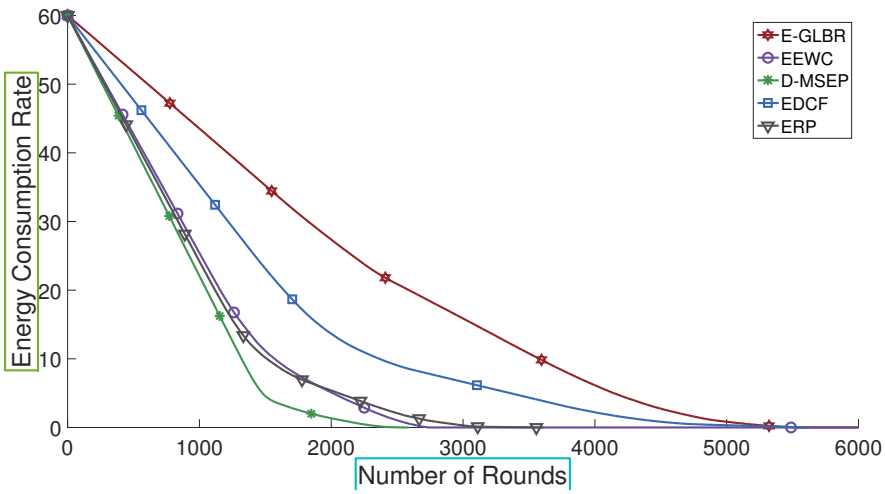

Figure 5: Total energy consumption Vs. number of rounds for $m = 0.2$ and $\alpha = 1$.

5.4. Network throughput

The quick death of sensor nodes has a significant effect on network throughput, when a huge percentage of nodes is disappeared, no cluster head would be selected and nodes data can not achieve the base station due to wide transmission range. The majority of network nodes in ERP and EEWC operate in a significant number of rounds, the duration between the death of the first node and the last node is short, in each round, the two protocols guarantee the selection of some cluster heads to ensure data delivery to the BS which leads to an important network throughput

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as shown in Fig.6. The algorithm of D-MSEP guarantees a significant period of network stability as mentioned in Fig.4, however, after the death of the first node, the remaining nodes deplete their energy so quick and 80% of nodes disappear in a few number of rounds, just 20% of nodes are operating in the next rounds, what explains the low throughput of this protocol. In EDCF network, the algorithm reaches a great network lifetime by preserving the energy of just 20% of the total number of network nodes, just 20 sensor nodes keep monitoring the whole field between the rounds 2618 and 5695, due to this low remaining number of nodes, the network falls into the hot spot problem where the sensor nodes are isolated from the communication with the base station and no suitable cluster head could be selected in all network areas, as a result the transmitted data from network nodes could not reach the base station. After the 2618 round, the maximum number of packets produced by the EDCF network is reached and no more packets is received by the BS.In our proposed E-GLBR, 50% of the sensor nodes operate for more than 4360, the network continues to form CHs in clustered regions for a large number of rounds compared to other protocols, furthermore, even if the number of nodes is reduced and no CH is selected, the gateway ensures the transmission of data from the nodes to the base station which keep receiving network nodes data from for more than 5000 rounds, the last packet is received in round the round 5525 where the last node is dead.

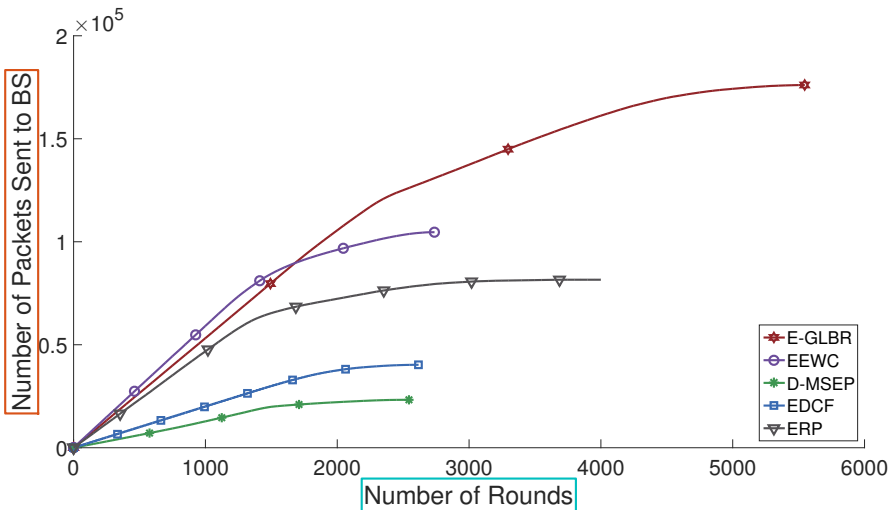

Figure 6: Network throughput Vs. rounds for for $m = 0.2$ and $\alpha = 1$.

5.5. Network Coverage

Network coverage can be considered as a metric to evaluate the QoS provided by a WSN, it presents the ability of a network to detect the maximum number of events or changes that may occur in the monitored area during the given network lifetime by a routing protocol. It is commonly assumed in the literature that the monitored surface $S(n_i)$ by a sensor node n_i is modeled by a circle of center n_i and a sensing range radius R_s [41]. We define the total covered surface at each round as follows:

$$S_t = \sum_{i=1}^{N} \phi_i * (\pi * R_s^2) \tag{13}$$

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where ϕ_i is a parameter that ensures that only alive nodes are taken into account when calculating the network coverage, it is represented as follows:

$$\phi_i = \begin{cases} 1 & \text{if } E(n_i) > 0 \\ 0 & \text{if } E(n_i) \leq 0 \end{cases} \tag{14}$$

We evaluated the performance of the discussed protocols in terms of network coverage by varying the percentage of advanced nodes m and the energy factor α, Fig.7 and Fig.8 demonstrate the average total area covered by each protocol for different values of m and α. we notice that, for all protocols, as the values of m and α increase, so does the coverage rate. The total covered surface of ERP and EEWC are almost the same in all scenarios, that result is due to the similar behavior of these networks in the previous discussed metrics including network lifetime and energy consumption rate. The total covered surface by D-MSEP and EDCF exceeds those of ERP and EEWC in all scenarios, generally, compared to ERP, D-MSEP and EEWC, the protocol EDCF covers wide surfaces in each round and in all values of advanced nodes percentage m and energy factor α, but the widest surfaces in all scenarios are covered by our proposed E-GLBR network,it increases the total area under coverage of EDCF by $25,23\%$, $28,37\%$, $26,28\%$, $27,41\%$ and $27,74\%$ for the energy factor values $\alpha = 0.5$, $\alpha = 1$, $\alpha = 1.5$, $\alpha = 2$, and $\alpha = 2.5$ respectively, and by 57%, 36%, 22%, 17% and 18% for the advanced nodes percentage $m = 0.1$, $m = 0.2$, $m = 0.3$, $m = 0.4$ and $m = 0.5$ respectively. The proposed E-GLBR protocol preserves the majority of nodes energy for the maximum number of rounds than all other protocols, in each round the number of alive nodes in our proposed network is greater than all other discussed network, each alive sensor node covers an area of $\pi * R_s^2\ m^2$. As the number of live nodes is large, so is the area covered by the network, which explains the significant results achieved by our proposed protocol.

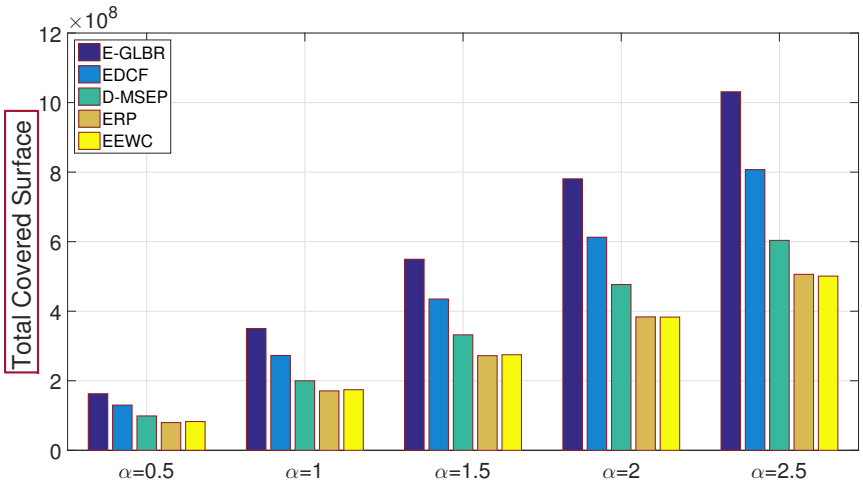

Figure 7: Total covered surface when α is varying.

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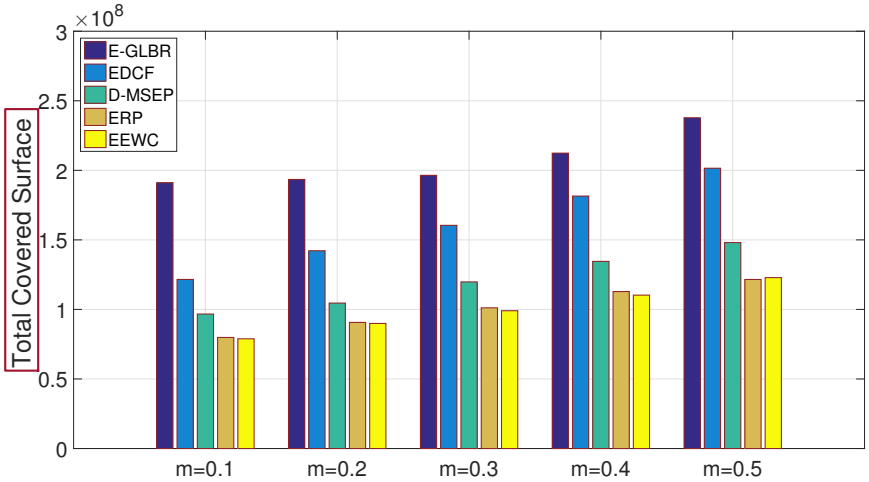

Figure 8: Total covered surface when m is varying.

6. Conclusion and future works

The principal purpose of a WSN is to monitor a physical area under the various constraints and limitations, this monitoring must satisfy a certain number of requirements, especially the stability period, network lifetime of the network and the quality of monitoring. Selecting adequate CH for data gathering has become a challenging task to achieve better network performance. In this research paper, the inclusion of residual energy, transmission distance, nodes density and CH energy consumption factors in the proposed E-GLBR fitness function has guaranteed the selection of optimal CHs. The simulation results has showed that the proposed algorithm achieves better performance in all network scenarios as compared to the discussed related methods. Thus, the enhancement of the optimized GA-based routing protocols was validated effectively. In future, we will try to implement other evolutionary algorithms in the proposed network model to enhance the provided QoS.

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